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Simulative investigation of network scalability over MANET routing protocols

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Abstract

The important feature that differentiates Mobile Ad-hoc NETworks (MANETs) to other Wireless or Wired Networks is the density of the nodes and mobility. MANET is a Network without Infrastructure, in which each node acts as a data sender, sink and router. Node density and Mobility are the key features that differentiate the MANET from the other wired and wireless networks. The self-configuring infrastructures without a network of mobile devices are connected via wireless connections. Any device on the MANET network can move to any direction and it frequently changes its connections with other devices. This network is an association of wireless nodes which may be dynamically configured at any time and from any location without utilizing the network infrastructure that is already in place. In multi-hop routing mode, nodes within radio range use other nodes as relays while distant nodes communicate directly over wireless links. The density of nodes or network scalability implies the quantity of nodes in the designed network scenario. The routing protocols are therefore designed for adapting the dynamic topology change while maximize its packet delivery ratio while minimize latency and throughput, with minimal number of packet losses and average jitter. Many researchers have been done the research work on MANET Routing Protocols. In this article, four routing protocols are used for the simulated scalable ad-hoc MANET implementations. The scalability refers to the variation in the number of nodes in a designed network scenario. For this research work, four scalable networks (25, 50, 75 100) are discussed. The five performance metrics are analyzed over four categories of node densities. The experimental results show that Ad Hoc Demand Vector (AODV) is 13.65% performed well in throughput than the other protocols; Dynamic Source Routing (DSR) is 12.66% performed well in goodput; DSR is 65% performed better than other protocols in the packet received; DSR is 10.3% performed better than other protocols in the received rate.

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Keywords: MANET; Random Waypoint; Node Density; AODV; DSR; DSDV; OLSR.

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1. Introduction

Ad-hoc, sensor, and automotive wireless multi-hop networks are just a few examples of important research areas in recent years. A collection of wireless mobile terminals known as MANET that is capable for dynamically establishing temporary networks without any assistance of the centralized management or fixed infrastructure. It is a collection of wireless mobile terminals that can dynamically construct a temporary network without centralized management or fixed infrastructure. The potential for its use in a variety of fields; MANETs have continued to garner attention over the past few years. In order to guarantee efficient operation when the number of nodes increases significantly, the overhead of the routing algorithms needs to be minimal and independent of the total no. of nodes in the MANET [1] [2]. MANET has attracted a lot of attention in recent years for potential applications in a variety of fields [3] [4]. The multi-hop Wireless network is ad hoc networks, sensor networks and vehicle networks that have been extensively utilized in the research. The MANET's mobility is a crucial parameter that affects the network performance very quickly. The decision about whether, where, and when to switch to another network is handled by mobility management. Coverage restrictions, bandwidth requirements, and other usages can influence forwarding decisions [5]. The two types of wireless networks cover a network that is not an infrastructure based and having an infrastructure. Fixed and wired gateways make up the fixed architecture of the infrastructure network. Another name for it is a cellular network. They have fixed base stations with cables connecting them to other base stations [6]. Its general architecture is shown in figure 1.
The infrastructure wireless LAN is the network of connect computers established on radio transmissions rather than wired connections. There is not the concept of movements of devices in wireless LAN. In ad-hoc networks, the devices can move from its fixed network and the topology makes it always dynamic rather than static.The performance of network varies when there is a change in the number of nodes (refers to scalability). It may be degraded due to high congestion or may be improved due to lesser number of nodes. So, scalability has the major impact on the performance of a network. The main **motivation** behind this research work to highlight the problems/ issues related to scalability over the network. The **contribution** of the work is to explore the impact of scalability on routing protocols on different performance metrics.

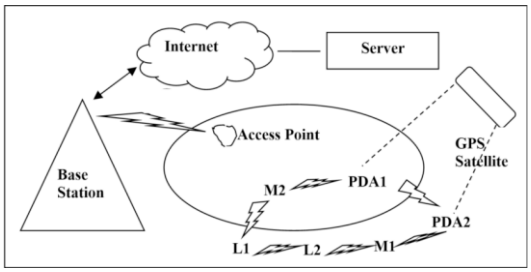

Fig. 1. MANET Architecture.

L*: Laptop
M*: Mobile
PDA*: Personal Digital Assistant
Literature Study and MANET Routing Protocols

1.1. Literature study

The relevant literature review comparison is shown in Table 1. The previous work related to this paper [7] dealt with the comparison of packet delivery ratio and packet loss metrics with respect to scalability parameter in designed MANET network scenario.

Table 1. Comparison of Literature Review.

Author,Year	Analysis of routing protocols	Merits	Demerits
Methaq A. Shyaa, Hassan M. Ibrahim, Mohammed Dauwed, Ahmed Meri, and Atheer Flayh [5] (2019)	DSR, DSDV, AODV, OLSR	Comparing packet delivery ratio and packet loss.	Increasing the node density in the scenario topology has bad impact on the network performance.

(Scopus indexed)			
Ankita A. Mahamune, M. M. Chandane (2021) [6]	AODV, DSR	Detection and prevention of malicious attacks.	It faces various network attacks in practice.
Pavle D. Bugarþiü, Marija Z. Malnar and Nenad J. Jevtiü [7] (2018)	AODV	Performance metrics are analyzed on NS-3 simulator.	There is the need of improvement in network performance.
Geetika Dhand and S.S.Tyagi [8] (2018) (Scopus indexed)	AODV, DSR, DSDV	The wired protocols are implemented in the wireless networks.	The large number of reply packets caused the control overhead.
Vinay Rishiwal, Sandeep Kumar Agarwal, Mano Yadav [9] (2016)	AODV	Analyzed the routing performance of AODV routing protocol.	Homogeneous MANETs has average energy consumption.
Ajay Kumar, Ravi Rai Chaudhary, and Satyam Kumar Sainy [10] (2016)	AODV, DSR	Performance metrics are investigated.	LAR1 has more delays than AODV and DSR.
Deepak Kumar Ashutosh Srivastava S. C. Gupta [11] (2012)	AODV, DSDV	The performance metric Delay and Throughput are compared on MANET Protocols using NS2.	AODV suffers in packet received and DSDV suffers in delay.
G. Jisha and Philip Samuel [12] (2011)	DSDV, AODV	The effect of varying node densities is investigated.	Some additional experimental work can be carried out.
Muhammad Shaffatul Islam, Nazmul Islam, Shah Alam, Adnan Riaz and Tanvir Hasan [13] (2010)	AODV	The performance of different voice codec under similar load conditions are analyzed.	Codecs with a high data rate and low data rate are not suitable for both scenarios.
N. Adam, M.Y. Ismail, J. Abdullah [14] (2010)	DSR, AODV	The performance metrics are investigated over MANET routing protocols using OPNET.	More packet loss and delay is depicted by AODV.
Ricardo de Oliveira Schmidt et al. [15] (2008)	OLSR	Variations in the scenarios are presented with the protocols.	Packet delivery rates are not improved.
Roxana Zoican [16] (2007)	OLSR, AODV	Scalability issues are addressed.	An open problem of multicasting is still existed.

1.2. MANET routing protocols

Because the arrangements of nodes in an ad-hoc network are varying, so MANET nodes must find it on their own because they are unaware of it. When a new node enters in an ad hoc network, it must first broadcasts its existence and arrival, along with listening capability from already existing mobile nodes. In **proactive routing**, each node contains one or more tables that hold the most recent information on the paths to any network node. A routing table is maintained by each node (for eg. DSDV, OLSR). A node only initiates a route discovery in **reactive routing** protocols when it wishes to send a packet to its destination. They do not regularly maintain or update their route tables to reflect the most recent route topology. A routing table is not kept up to date by a node(for eg. DSR, AODV). The reactive and proactive features of the table maintenance mechanism and the route discovery mechanism are incorporated into the design of the **hybrid routing** protocols, respectively. The hybrid protocol works well in the Ad-Hoc networks with many nodes [14]. The selection of a route depends upon the selection of a routing protocol. If it is a proactive routing protocol (DSDV, OLSR) then the routing table is approached on the changes of topology (which is already made and updates periodically). If it is reactive routing protocol (AODV, DSR), then a new route request message will be broadcasted in the network to establish a new route on the topology changes. In this research work, the experiment is carried out on DSDV, DSR, AODV, and OLSR.

*1.2.1.Ad-hoc On-Demand Distance Vector (AODV)*It falls under the genre of Reactive Protocols, and it does not require movable nodes to track journeys to locations where rest of the movable nodes is not communicating. Instead, it just makes route requests when necessary. AODV ensures that a route does not contain loops by using a sequence number to indicate how recent it is. AODV protocol only creates routes between nodes when requested by source nodes. The routes will be maintained for as long as the sources demand them. They also connect multicast group members by constructing trees. Networks in AODV remain silent until connections are established. Requests for connection are broadcast by network nodes that require connections. The remaining AODV nodes record the node that requested a connection and forward the message. As a result, they design a series of indefinite routes that lead back to the requesting node. A backward message is sent to the requesting node through temporary routes by a node that holds a route to the desired node and receives such messages. The route with the fewest number of hops through other nodes is taken by the node that initiated the request. After some time all the entries that are not used in

the routing table that must be reused. The procedure is repeated that if any link breaks and the routing error occur then it is sent back to its transmitting node. The route request is a node that initiates the sending or receiving of a packet but does not know how to get there. It initiates the route discovery process by sending a Route Request (RREQ) multicast message. Before the message reaches the destination node, neighbors keep track of where it came from and forward it to their neighbors. A destination node responds with a Route Reply (RREP), which follows the RREQ's route back to the source in the Route Reply. Intermediate nodes create direct paths when the RREP returns to the source. If the intermediate node knows the route to the destination, it can send an RREP in response to the received RREQ request, allowing the nodes to follow a predetermined route. Source-destination matching begins when the RREP reaches the source and a header is set. Because it is a reactive protocol, a path failure in AODV generally has less overhead than a proactive protocol. The Route Error (RERR) message is sent by the node that detected a path break if messages cannot be sent. The message turns to different centers. The RERR message indicates the target is out of range. The path is disabled by nodes receiving messages. The routing tables with one entry for each destination with which it communicates must be kept by each AODV nodes. The route entries keep tracks of a number of the fields including the Next Hop, Hop Count, and Destination IP Address. The nodes are broadcasts at a RREQ packets that are locate at a route to its destination by utilizing AODV. The route request for the destination includes the IP address of the source nodes and the current sequence number to the Broadcast ID and the very new sequence number. The Destination Nodes unicasts a RREP packet along with the reverse paths that are organized at the intermediate node during the process of route discovery after receiving RREQ. In the event of a link failure, both the Source and Destination nodes receive a packet indicating a RERR. The Sequence Numbers allow the source nodes to always locate new and legitimate routes [15]. The route finding process of AODV is represented in figure 2.

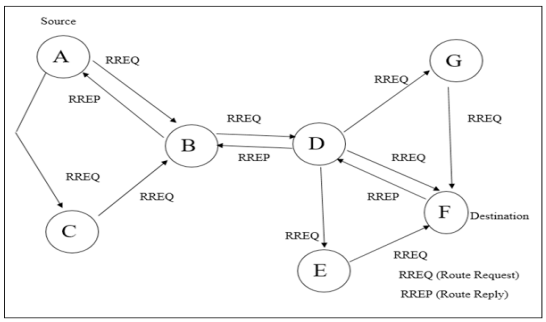

Fig. 2. Route finding process in AODV.

*1.2.2. Dynamic Source Routing (DSR)*The DSR, like AODV, produces the destination whenever the source node requests the route. DSR uses a source routing method. When using source routing, the source must travel each hop on the route to the destination. The route cache stores a known route for each node. If the requested path cannot be found in the path cache, a path discovery is performed. DSR employs source routing, which accurately specifies the hops via which each packet should transit. As a result, every packet's header must provide the hop order. This has the disadvantage of requiring each packet to carry additional routing overhead. However, the ability of intermediate nodes to learn routes from the source routes in packets they receive is a considerable advantage. Because determining a route usually requires a lot of time, bandwidth, and energy. This is an excellent argument to employ source routing. Source routing also has the advantage of not requiring packets to carry current routing information in the intermediate. Finally, because a single node chooses the full path rather than hop-by-hop, routing loops are easily avoided. Figure 3 depicts the route-finding process of DSR.

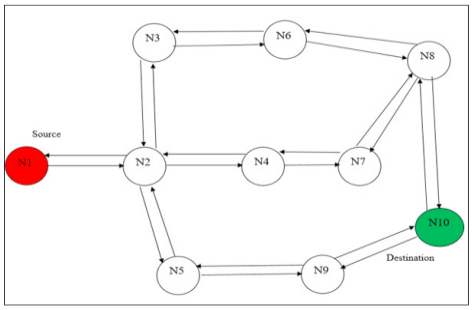

Fig. 3. Route finding process in DSR.

*1.2.3. Optimized Link State Routing (OLSR)*OLSR utilizes three routing mechanisms: 1) Send out "Hello" messages on a regular basis to keep track of neighbors. 2) To prevent packet flooding, use Multipoint Relay (MPR), and 3) Select a route using the shortest path first technique. Each node has two hop neighbors from which it can be reached. Following that, only messages transmitted by nodes that have chosen it as an MPR are rebroadcast by other nodes. Each node has a partial network topology graph because this approach significantly minimizes the overhead of broadcast control. Each MPR-selected node announces its presence by sending Topology Control (TC) signals to its selector set. TC messages contain the originating node's address as well as its MPR selection set. Once the source node has access to the routes, it uses the shortest path first approach to select the best route. Figure 4 depicts the route finding process for OLSR [6].

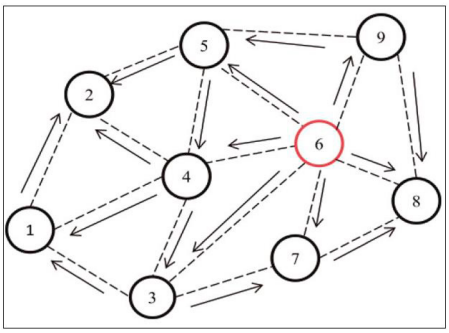

Fig.4. Route finding process in OLSR.

*1.2.4. Destination Sequenced Distance Vector (DSDV)*A major issue with wired distance vector routing is addressed by the proactive routing protocol, DSDV. It requires each mobile station to notify its current neighbors of the existence of its own routing table. The advertisement needs to be shown frequently enough so that each mobile computer can always located the more mobile computers because the entries in this list can change quickly over time. In addition, every mobile computer agrees to transmit data packets to other computers upon request. The DSDV protocol ensures that there are never any loops on any path to any destination. DSDV is the MANET table-driven routing algorithms that are very much used. A routing table is kept by the DSDV Routing Protocol. It contains lists of all possible destinations, the number of hops required to get there, and the sequence number that the destination node assigns to each one [12]. The route finding process of DSDV is represented in figure 5.

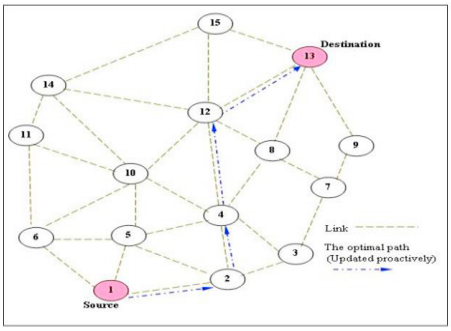

Fig.5. Route finding process in DSDV.

As per the research covered in this area, the following **objectives** have been indentified:
1. Perform simulative experiments over different scalable MANET networks.
2. Analyze the performance based on different Quality of Service (QoS) performance metrics.

2. Proposed MANET model and performance metrics

2.1 Proposed MANET model

Figure 6 represents the proposed MANET model for the complete simulative experiment. In this diagram an Input and the wireless link (i.e., IP address) are present. The wireless links are connected to different routing protocols. There are four routing protocols (AODV, DSDV, OLSR and DSR) acting in this environment. All protocols are performing on different node densities by selecting its respective protocol option value (1=AODV, 2=DSR, 3=OLSR, 4=DSDV).

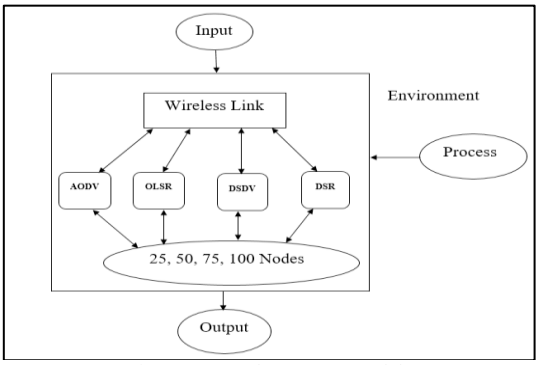

Fig.6. Proposed MANET Model.

Pseudo Code
1. Take manet.cc file as an input.
2. Set the wireless protocol (i.e. 802.11b).
3. Select the Protocol (1=AODV, 2=DSR, 3=OLSR, 4=DSDV).
4. For (i=25; i<=100; i=i+25)
{
Execute the manet.cc file & Take the outcome.
}
5. If want to take result of next protocol then go to step 3.
6. Else exit.

Fig.7. Pseudo Code of Flow Process.

The step-by-step pseudo code is depicted in figure 7. For experiment work, manet.cc file is taken as input in NS-3 software. The value of wireless protocol is 802.11b. The selected parameters are simulated along with a chosen

routing protocol. Each routing protocol is simulating on chosen category of node density at a time. This work is incorporated under for loop. For selected parameters, the manet.cc file is executed and shows its output. For next simulation, the control goes back to step 3 again; otherwise it terminates the simulation.

2.2 Performance metrics

Response time, throughput, resource utilization, error rate, and scalability index are common metrics. In this paper, Quality of Service performance metrics named as throughput, goodput, packet received and receive rate are measured. The metrics are compared across four MANET routing protocols. The experiment time utilized to find the results is set to 100 seconds. The area is set to be 1500 and 300 meters. Each routing protocol requires scalability of 25, 50, 75, or 100 mobile nodes. The node positioning is random. The mobility speed is 10 m/s. The number of sinks is 10. The Random Waypoint mobility model is used for the movements of nodes. Table 2 lists the network's parameters.

Table 2. List of Parameters.

Parameter Name	Parameter Value	Detail
Time	100 seconds	The execution time of the designed scenario.
Area	1500*300m^2	The row-wise and column-wise dimensions.
Number of mobile nodes	25, 50, 75, 100	The four scalable network scenarios.
Placement	Random	A node position in random manner.
Speed	10m/s	The speed of a node is 10m/s.
Mobility Model	Random Waypoint	The movement pattern of a node.
Path-finding	AODV, DSR, OLSR, DSDV	The path-finding mechanism in the designed scenario.

a) Throughput

Throughput in MANET can be defined as the rate at which data is successfully transmitted over the network. It is usually measured in bits per second (bps) or packets per second (pps). The formula to calculate throughput in a MANET can be expressed as:

$$\text{Throughput} = (\text{Total number of successfully received packets}) / (\text{Total Simulation Time}) \quad (1)$$

b) Goodput

Goodput in MANET is the measure of useful transmitted data over the network, excluding protocol overhead, retransmissions, and other non-payload data. It reflects the actual application-level throughput experienced by the user. The formula for calculating goodput is:

$$\text{Goodput} = (\text{Total number of useful bits received}) / (\text{Total Simulation Time}) \quad (2)$$

c) Packet Received

It counts the total number of data packets that successfully reach their intended destination over a given period. The formula for calculating the total number of packets received in a MANET is straightforward as:

$$\text{Total Packets Received} = (\text{Total Packets Sent}) - (\text{Total Packets Lost} \quad (3)$$

d) Received Rate

The received rate in a MANET is a measure of how many packets are successfully received over a given time period. It is typically expressed in packets per second (pps). The formula to calculate the received rate is:

$$\text{Received Rate} = (\text{Total Packets Received}) / (\text{Total Simulation Time}) \quad (4)$$

3. Results analysis and discussions

The performance metric throughput is presented in figure 8. As shown in the figure, DSR performed better when number of nodes are 25 and 50 than the other routing protocols. But its performance degraded when number of

nodes are increased to 75 and 100. OLSR is consistently played better in all categories of scalable scenarios except 75 nodes. DSR allows multiple routes to any destination and allows each sender to select and control the routes used in routing. The performance of AODV is on the first position in the category of 75 nodes. AODV does not put any additional overheads on data packets as it does not make use of source routing. DSDV performed better upto 75 nodes in increasing order but its performance degraded when it reaches to 100 nodes category. DSDV is quite suitable for creating ad-hoc networks with small number of nodes. OLSR is consistently improving with the number of nodes increasing. OLSR increases the protocol's suitability for an ad-hoc network with the rapid changes of the source and destination pairs. AODV is 13.65% performed well in throughput better than other protocols.

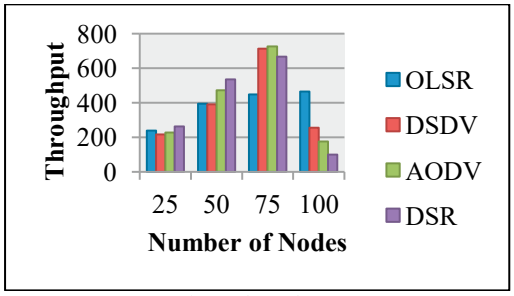

Fig.8. Throughput.

The comparative analysis of performance metric goodput is shown in figure 9. In the overall comparison, DSR performed better in the all categories of scalable scenarios than the other routing protocols. DSR has the quick path recovery mechanism in case of broken links which helps to achieve more goodput. After DSR, DSDV performed well on second position. DSDV only stores the best and shortest path by using the Bellman_Ford algorithm which helps to get more goodput. AODV performed better on third position in all categories on scenarios. AODV performed well in highly dynamic applications and it contains information about the active routes only. The goodput value for OLSR routing protocol is lower most in all categories of network scenarios. The poor performance of goodput for OLSR is due to its used storage capacity about the routes which may never be used which is the wastage of resources. Moreover, the performance of network in goodput metric is degraded when the number of nodes are increased in the scalable scenario. DSR is 12.66% performed well in goodput as compared to other routing protocols.

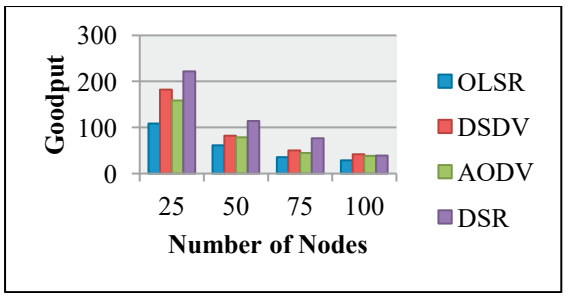

Fig.9. Goodput.

The comparative analysis of performance metric packet received is shown in figure 10. Both routing protocols, OLSR and DSR performed well in all categories of scalable scenarios. OLSR provided better results because it does not require that the link is reliable for the control messages. Moreover, the delivery does not have to be sequential. Whereas, the reason of better performance of DSR is that it supports multipath routing and quick path recovery mechanism in the case of broken links. After that, AODV is consistently performed well in all categories of scenarios. The reason of moderate performance of AODV is the support of both kinds of packets transmissions: unicast as well as multicast. Whereas, DSDV is performed as the lowest most routing protocol in packet received metric due to the non-support of multipath routing and wastage of bandwidth in route advertisement periodically. DSR is 65% performed better than other protocols in the performance metrics packet received.

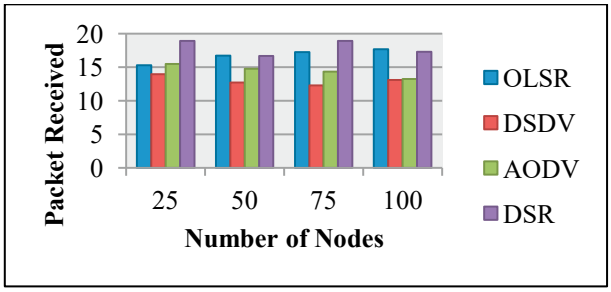

Fig.10. Packet received

The comparative analysis of performance metric received rate is shown in figure 11. In the overall comparison, DSR is performed well in all categories of scalable scenarios. It is because of the route creation only when it is required and the nodes utilize this route information efficiently which results in less overhead and collision. After DSR, OLSR is performed better than the other routing protocols due to less end-to-end delay and it worked with fewer headaches than other routing protocols. AODV is performed well on third position. It is due to its quick response to the dynamic changes in the topology which affects various routes. Whereas, DSDV is performed worst in all categories of scalable scenarios. It is due to regular updates of its routing tables which uses more battery power and a small amount of bandwidth. DSR is 10.3% performed better than other protocols in the performance metric Received Rate.

The numerical analysis of throughput and goodput is depicted in table 3 and numerical analysis of packet received and received rate is shown in table 4.

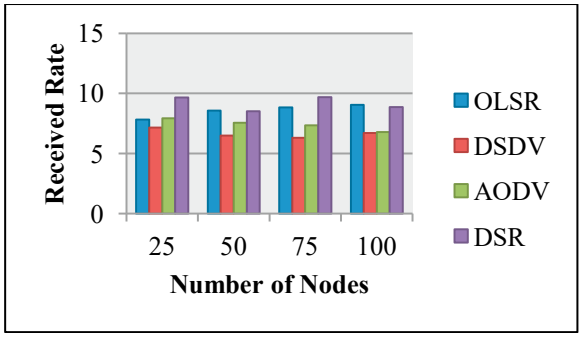

Fig.11. Received rate.

Table 3. Numerical Analysis of Throughput and Goodput.

Number of Nodes	Throughput				Goodput			
	OLSR	DSDV	AODV	DSR	OLSR	DSDV	AODV	DSR
25	238.7288	395.3564	448.782	465.0504	108.5816	182.0928	158.5584	221.9948
50	217.4692	391.3506	713.98	254.8808	61.62	82.1578	78.9796	114.2336
75	228.5141	472.9956	725.9229	175.4363	35.44267	49.93907	45.0136	77
100	262.9589	535.7804	667.2608	99.0215	28.61	42.2336	38.3876	39.0397

Table 4. Numerical Analysis of Packet Received and Received Rate.

Number of Nodes	Packet Received				Received Rate			
	OLSR	DSDV	AODV	DSR	OLSR	DSDV	AODV	DSR
25	15.27	13.98	15.48	18.9	7.81824	7.15776	7.92576	9.6768
50	16.74	12.7	14.77	16.66	8.57088	6.5024	7.56224	8.52992

75	17.25	12.31	14.33	18.94	8.832	6.30272	7.33696	9.69728
100	17.66	13.12	13.24	17.29	9.04192	6.71744	6.77888	8.85248

4. Conclusions and future scope

Wireless networks are computer networks that are not connected by wires. The wireless networks are used in such kind of applications where a business cannot bear the cost of installing the cable connections among the various communicating devices. The radio waves that are fundamental components of wireless systems and they are utilized at the physical level of a network's structure. MANET is a network that operates without strict top-down network management and has many free or autonomous nodes that can arrange themselves in many different ways. These nodes are typically mobile devices or other mobile pieces. MANET is considered to be as the base of further studies like VANET, FANET and SANET. The scalability issue needs to be addressed in all such kind of studies. This study may help the researchers to do research work in the relevant fields. The results showed that AODV is 13.65% performed well in throughput better than other protocols; DSR is 12.66% performed well in good put; DSR is 65% performed better than other protocols in the Performance Metrics Packet Received; DSR is 10.3% performed better than other protocols in the Performance Metrics Received Rate. Now, the research work is restricted up to 100 nodes only. In future, simulative study up to 300 nodes and more can be performed. Some other performance metrics like average end to end delay, average jitter value, energy efficiency, and threshold value can also be analyzed. The future scope is restricted and highly congested scenario up to for 600 numbers of nodes. Also mobility speed parameter can be merged with the node density. The implemented research work is covered at macroscopic level; the microscopic analysis may be covered in the future.

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